

IRFAN BASHA

DEVOPS ENGINEER | EKS | KUBERNETES | TERRAFORM | CI/CD
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LINKEDIN GITHUB PORTFOLIO

Professional Summary

DevOps Engineer with hands-on experience in AWS, Kubernetes, and Terraform. Architected a three-tier production application on EKS achieving 99.9% uptime and 40% cost optimization. AWS Certified Solutions Architect – Associate.

EXPERIENCE

DevOps Engineer Intern - PearlThoughts (Remote)

Dec 2025 – Jan 2026

- Took ownership of deploying a Strapi (headless CMS) application end-to-end: built the AWS infrastructure, containerized the app, deployed to Kubernetes, and automated the release pipeline
- Authored Terraform modules to provision AWS infrastructure (VPC, Security Groups, IAM, EKS, ECR), applying Infrastructure-as-Code principles for reproducible environments
- Containerized the Strapi application with Docker, pushed images to Amazon ECR, and deployed workloads to Amazon EKS
- Gained hands-on experience with AWS Fargate (ECS) for serverless container deployments, removing the need to manage underlying EC2 infrastructure
- Built GitHub Actions CI/CD workflows automating build, test, image push to ECR, and rollout to EKS on every merge
- Configured AWS networking and IAM policies following least-privilege principles
- Troubleshoot deployment issues, contributed to infrastructure optimization, and supported application monitoring in collaboration with the development team

PROJECTS

Three-Tier E-Commerce Platform Deployment on AWS EKS :

- Architected enterprise-grade cloud infrastructure deploying a complete three-tier application (ReactJS, NodeJS, MongoDB) on Amazon EKS with 99.9% uptime using AWS Load Balancer Controller and auto-scaling, serving 50K+ daily requests
- Implemented Terraform IaC reducing provisioning from 4 hours to 15 minutes, and deployed Jenkins CI/CD pipeline with ArgoCD GitOps, cutting deployment time by 70% and manual intervention by 80% across 12+ microservices
- Built production-ready observability stack (Prometheus + Grafana + custom dashboards) across 50+ services, reducing MTTR from 3 hours to 25 minutes and enabling proactive incident detection
- Optimized infrastructure costs by 40% through efficient resource management, automated cleanup processes, and strategic design, saving \$12K/month in AWS spend

KubeHealer - AI-Powered Kubernetes Auto-Remediation System

- Built KubeHealer, an AI-powered Kubernetes auto-remediation system that autonomously scans, diagnoses, and fixes pod failures (CrashLoopBackOff, OOMKilled, ImagePullBackOff), reducing manual intervention and MTTR
- Leveraged Temporal workflow orchestration to achieve 100% crash-proof execution with automatic retry mechanisms, state recovery, and complete audit trails for all remediation actions
- Integrated Groq LLM (Llama 3.3 70B) for intelligent root cause analysis, enabling the system to identify issues like image typos, resource limits, and configuration errors with natural language explanations
- Demonstrated cost optimization by using Groq's free API tier instead of commercial alternatives, while maintaining high-quality diagnosis tool-calling capabilities for production-grade workflows

Technical Expertise

- **Cloud & Infrastructure:** Amazon Web Services (AWS), Microsoft Azure | Terraform, Ansible | IAM & Security
- **Container Orchestration:** Docker, Kubernetes (K8s), Helm, Amazon EKS, KEDA | Auto-scaling, HPA
- **CI/CD & GitOps:** Jenkins, Azure Pipelines, ArgoCD, GitHub, GitHub Actions, Git | Pipeline as Code
- **Monitoring & Observability:** Prometheus, Grafana, EFK Stack, Jaeger, CloudWatch | Alerting, Log Analysis, SLO/SLI
- **AI & Generative AI:** Claude CLI, Agentic AI, RAG (Retrieval-Augmented Generation) | LLM Integration, AI Workflow Automation
- **Scripting & Programming:** Python, Shell, SQL, YAML | Automation Framework Development, Flask
- **Networking:** TCP/IP, DNS, Load Balancing, VPC, Security Groups | netstat, ping, traceroute, nc, tcpdump, SSH/SCP
- **Operating Systems:** Linux Administration, Ubuntu
- **Soft Skills:** Problem-solving, Analytical Thinking, Communication & Collaboration, Continuous Learning

Certification



- **AWS Certified Solutions Architect – Associate**

Credential URL - www.credly.com/badges/8336f37b-d694-499c-812c-06e86db040c7

Validity - July 21, 2025 - July 21, 2028

Education

Aalim muhammed salegh college of engineering
B.E. Computer Science Engineering | 2021 – 2025